Chapter 5: Prime Time

Worksheet 5: Primes, Coprimes, and Divisibility Explorations

Learning Objective: Students will identify prime numbers, understand coprime numbers, explore the Sieve of Eratosthenes, and apply prime number concepts to factorisation and problem-solving.

Worked Example

Are 14 and 15 coprime?
Solution: $14 = 2 \times 7$; $15 = 3 \times 5$. They share no common factors except 1.
$\therefore$ HCF(14, 15) = 1. Yes, they are **coprime**.

Find the number of primes from 1 to 20 using the Sieve: 2, 3, 5, 7, 11, 13, 17, 19 $\Rightarrow$ **8 primes**.

Questions 1–5: Easy / Recall

Q1. List all prime numbers less than 30.

Q2. Is 1 a prime number? Why?

Q3. What is the only even prime number?

Q4. Are 8 and 15 coprime? Show working.

Q5. How many prime numbers are there between 10 and 20?

Questions 6–10: Basic Application

Q6. Use the Sieve of Eratosthenes to find all primes up to 50. (Teacher Verification Required for the sieve)

Q7. Write 36 as a sum of two prime numbers in two different ways.

Q8. Are 25 and 36 coprime? What is their HCF?

Q9. Twin primes are pairs of primes that differ by 2. List three pairs of twin primes.

Q10. Every even number greater than 2 can be written as a sum of two primes (Goldbach's Conjecture). Verify for 28, 36, and 50.

Questions 11–15: Practice and Skill Building

Q11. How many factors does a prime number always have?

Q12. Find three numbers less than 100 that have exactly 3 factors. What type of numbers are these?

Q13. A prime number $p > 5$. Is p always odd? Is p always ending in 1, 3, 7, or 9?

Q14. Find the prime factorisation of $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$.

Q15. Which number between 90 and 100 has the most factors? List them.

Questions 16–18: Higher Order Thinking

Q16. A *perfect number* equals the sum of its proper factors. Verify that 6 is perfect $(1 + 2 + 3 = 6)$. Check if 28 is perfect.

Q17. How many prime numbers between 1 and 100 end in the digit 3? List them.

Q18. If p and q are distinct primes, how many factors does $p \times q$ have? How many does $p^2 \times q$ have? Generalise.

Questions 19–20: Real-Life Applications

Q19. Cryptography (the science of secret codes) uses very large prime numbers. A simple code multiplies two primes: $p = 7$ and $q = 11$ to get $N = 77$. If you know only $N = 77$, how would you find p and q? Try $N = 143$.

Q20. In a school of 120 students, teams must be formed with equal members. A prime team size means no further equal subdivision is possible. Which prime team sizes divide 120 exactly?

Reflection Question: Why are prime numbers described as the “building blocks” of all whole numbers? Explain with an example.